Machine Learning Application for Credit Card Fraud Prediction and Security Risk Monitoring Model

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Abstract
In light of the rising prevalence of digital payments and the associated surge in small-value fraud, this paper develops a comprehensive framework for credit card fraud detection and transaction risk monitoring. Leveraging a real-world dataset sourced from Kaggle, our model suite includes both Supervised (Decision Tree, Random Forest, XGBoost) and Unsupervised (Isolation Forest, Autoencoders) Machine learning algorithms, with performance evaluated using precision, recall, and false positive rates. We also introduce a hybrid risk scoring system that integrates model-based predictions with domain-driven rules, enhancing interpretability while increasing sensitivity to fraud signals. The rule-based model demonstrates exceptional concentration of actual frauds in top-risk categories, though at the cost of flexibility. This multi-pronged approach highlights the trade-offs between model accuracy, interpretability, and operational feasibility, offering actionable insights for financial institutions aiming to implement scalable, real-time fraud detection systems.

1. Introduction

1.1 Motivation and Relevance

The importance of credit card fraud detection is underscored by its profound financial and strategic implications for both financial institutions and consumers. This challenge can be examined through four key lenses derived from a structured problem-solving framework: (i) Credit card profitability, (ii) Market growth and trends, (iii) Fraud risk in retail, and (iv) Customer retention.

Credit cards account for approximately 15% of issuer profits through interest and fee-based income, positioning them as both a vital revenue stream and a high-value target for financial crime. The urgency of addressing fraud is further intensified by shifting global consumer behavior—credit card spending is projected to grow at a compound annual growth rate (CAGR) of 8.5%, fueled by the rapid expansion of e-commerce and advances in digital payment technologies.

At the same time, fraud is escalating, with low-value fraudulent transactions contributing to an estimated $43 billion in global losses. This growing threat highlights the need for robust fraud prevention strategies. Moreover, customer retention is increasingly tied to a bank's ability to detect and respond to fraud swiftly. Delayed action not only results in financial loss but also damages consumer trust—making early fraud detection a critical element of both risk management and long-term customer loyalty.

2. Data Processing

2.1 Data Extraction

The dataset used in this study was sourced from Kaggle and comprises anonymised credit card transactions records, structured across multiple dimensions relevant to fraud detection.

2.1.1 Key features Extracted

The dataset includes various transaction-level features that can serve as input variables for training the fraud prediction model. These features, detailed in Table 1, are instrumental in capturing behavioral patterns associated with fraudulent activity.

Features	Rationale
Transaction timestamp	Temporal information provides insights into behavioral patterns and periodic anomalies that are often characteristic of fraud.
Merchant category	Merchant categories serve as risk profiling features, as some merchant types are more susceptible to fraud due to volume or ease of access.
Transaction amount	Fraudulent activity frequently involves unusually large or small amounts, deviating from typical consumer behavior.
Merchant and customer latitude and longitude	Geolocation features such as merchant and customer latitude/longitude facilitate the computation of distance-based features, which can signal spatial anomalies.
Customer date of birth (DOB)	Demographic information like the cardholder's date of birth allows for age-based segmentation, which supports profiling and detection of outlier behaviors in specific age groups.
Fraud binary labels	This column serves as the binary target variable for supervised learning, labeling transactions as fraudulent or not. These extracted features collectively form the foundation of our model, supporting both statistical inference and predictive performance.

Table 1. Key Features for Prediction Model and Analysis

2.2 Data Cleaning

2.2.1 Target Variables

The dataset contained mixed data types, including numeric, string, and timestamp-like entries, which presented challenges for binary classification. To address this, we standardized the input by encoding string-based values. This preprocessing step ensured consistency and compatibility with the binary classification model.

a) **Temporal Features**

From the raw timestamp data, we split into date, hour, and day of the week. These enabled the identification of temporal patterns typical of fraudulent activity. Furthermore, we created a binary variable for weekends, to differentiate transactional activities on weekdays and weekends. Some records lacked timestamp information, resulting in null values. To address this, a missing time flag was introduced to explicitly mark these cases. This allowed the model to treat missingness as a potential signal rather than a source of bias during training.

b) **Age of credit card holders**

The age of credit card holders was derived from the date of birth (DOB) field. For records with missing DOB values, we applied mean imputation, replacing them with the average age to preserve dataset completeness and minimize potential bias. Unlike the timestamp feature, we did not introduce a missing value indicator for DOB, as this feature was intended for descriptive insights during EDA, rather than serving as a predictive variable in the modeling pipeline.

c) Geospatial Distances

To capture spatial anomalies often linked to fraudulent transactions, we computed the great-circle distance between the transaction location and the merchant location using the Haversine formula. This feature helps identify suspiciously large deviations from typical geographic behavior.

2.2.2. Categorical Variables

Categorical features such as merchant, category, city, state, and job were transformed to numeric format using label encoding. This approach is particularly well-suited for tree-based algorithms like XGBoost and Random Forest, as it preserves the integrity of category labels without introducing artificial ordinal relationships.

2.3 Data Imbalance

2.3.1 Handling Class Imbalance

Class imbalance is a well-known challenge in fraud detection tasks, where the minority class—fraudulent transactions—constitutes only a small portion of the dataset. In our case, just 12.77% of the observations were labeled as fraudulent (Class 1), while 87.23% represented non-fraudulent transactions (Class 0). This significant imbalance can bias the model toward predicting the majority class, potentially resulting in deceptively high overall accuracy but poor recall for the minority class. Such a scenario undermines the primary objective of fraud detection: identifying rare but critical fraudulent events.

2.3.2 Synthetic Minority Oversampling Technique (SMOTE)

To mitigate the effects of class imbalance, we employed the Synthetic Minority Oversampling Technique (SMOTE). Before applying SMOTE, median imputation on training features was performed to address missing values and ensure a complete dataset for oversampling. SMOTE generates synthetic samples for the minority class by interpolating between existing examples, enhancing the model's ability to learn meaningful patterns associated with rare events—without resorting to simple duplication. It is important to note that SMOTE was applied only to the training set to avoid data leakage and preserve the integrity of the evaluation process. This approach was designed to improve key performance metrics such as recall and F1 score, thereby strengthening the model's effectiveness in detecting fraudulent transactions.

2.4 Exploratory Data Analysis (EDA)

To gain preliminary insights into the transaction dataset, we conducted a series of exploratory data visualizations, which are presented in the Appendix. Each slide in the appendix (Figures A1 to A3) contains two diagrams that highlight distinct behavioral and statistical patterns relevant to fraud detection.

In Appendix Figure A1, the first diagram shows the transaction amounts are heavily skewed to the right, where most fraudulent transactions take place during the later part of the day. The other diagram which focuses on weekday & weekend transactions, shows that the highest fraud rate occurs during weekdays and fraudsters target periods of low activity, due to merchants possibly being less vigilant.

In Appendix Figure A2, it shows the fraudulent transactions by merchant category, with e-commerce and offline groceries having the highest rate of fraud. It can be inferred that fraudsters tend to target online transactions as there is no physical paper trail. Furthermore, higher frequency transactions make it harder to detect fraud.

In Appendix Figure A3, it shows the range of amounts for fraudulent transactions, and it concludes that the amounts fall in the 200-500 & 500-100 range mostly. Furthermore, in the other diagram, both fraud and non-fraud transactions are concentrated within the age of 60 years. One can infer that fraudsters prey on the elderly as they are easy targets.

3. Model Building and Implementation

3.1 Supervised Learning: Fraud Label Prediction Model

In developing credit card fraud prediction and security risk monitoring systems, we selected Decision Tree, Random Forest Classifier, and XGBoost as our supervised learning models. Prior to the model construction, the dataset underwent several preprocessing steps to ensure the quality of input features. We began with feature selection, identifying four primary feature groups to be used as predictors (X variables): transaction amount and location, time, demographics, and merchant category (Table 2). Conversely, the binary fraud label (i.e., 'is_fraud') was designated as the target variable (Y).

Feature group	**Variables (X)**	**Definition**
Amount and location	Transaction amount (in $) Transaction amount bin Distance to merchant City State City population	Monetary amount of transaction Range of transaction amount (i.e., < $50) Distance between transaction and merchant City where transaction occurred State of the transaction Population of the corresponding city
Time	Hour Day of week Weekend indicator Missing time flag indicator	Time of day (i.e., 0am to 12am) Day of the week (i.e., 0 to 6 for Mon to Sun) Boolean type (i.e., 1 = weekend, 0 = weekday) Boolean type (i.e., 1 = missing, 0 = available)
Demographics	Age Job type	Cardholder age Cardholder occupation
Merchant profile	Merchant category	Industry or service type of the merchant

Table 2. Predictor Variables of Model

To ensure model robustness and class balance, the dataset was split into 70% training, 15% validation, and 15% testing using stratified sampling. For missing values in the predictor variables, we used median imputation, which is a robust method that avoids distortion from outliers and ensures compatibility with most machine learning models.

Furthermore, our extracted raw data suffers from high class imbalance issues with only 12.77% of datasets representing the minority or fraud label classes. To address class imbalance issues which are common in real fraud cases (i.e., rare events), we applied the Synthetic Minority Over-sampling Technique (SMOTE) on the training set. SMOTE generates synthetic fraud samples by interpolating between existing minority class instances, rather than duplicating them. This helps the model learn more generalized decision boundaries and enhances its ability to detect fraud cases without overfitting and bias toward the dominant class. Finally, to ensure faster model convergence and equal contribution of features, we applied standard scaling process using StandardScaler to the input features given those variables have different metrics.

3.1.1 Decision Tree

The Decision Tree is one of the most intuitive and interpretable supervised machine learning models. It classifies data by recursively splitting it into smaller subsets based on features that best separate fraudulent from non-fraudulent transactions. At each decision node, the model will select a feature and a threshold – such as transaction amount, time of the day and/or merchant category that helps in maximising the purity of the resulting groups. Through this structure, a rule might state that an online transaction exceeding $500 between 1 a.m. and 4 a.m. should be flagged as high-risk. These clear “if-then” rules make the decision tree highly interpretable, which is a critical advantage in financial applications where transparency and explainability are required to justify why a transaction is blocked or approved.

Unlike black-box models such as neural networks or ensemble methods, every decision in a decision tree is completely transparent and traceable, making it suitable for regulatory environments application. In addition, Decision Tree is able to naturally capture nonlinear relationships between features without requiring feature scaling or transformation. This is particularly valuable in fraud detection where subtle and complex interactions—for instance, between merchant category, transaction time, and geographic distance—may jointly indicate suspicious behavior.

In our implementation, we trained the model on a SMOTE-balanced training set to mitigate class imbalance and ensure effective learning from the minority fraud class. To identify the model’s performance, we applied standard metrics on both validation and test sets using: accuracy, precision, recall, F1 score, and ROC-AUC. Based on Table 3, the model showed strong prediction performance on both validation set and test set. The high recall scores imply that the model is able to capture most fraud cases while avoiding many false alarms, represented by high precision scores with overall accuracy of 96%.

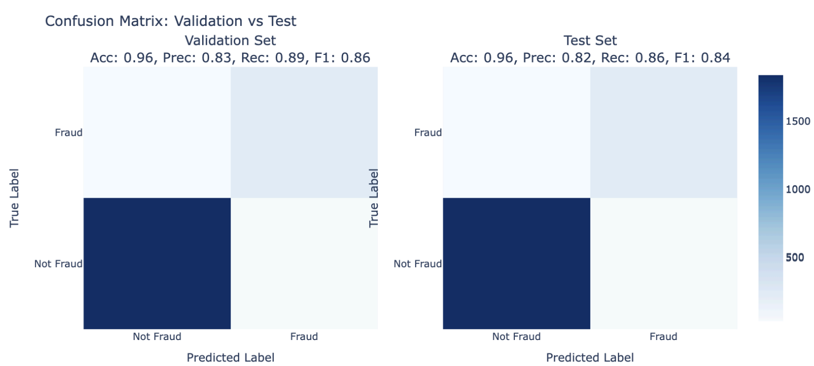


Fig 1. Confusion Matrix of Decision Tree Model for Validation and Test Set

Label	Precision	Recall	F1 Score	Accuracy
Validation Set				[redacted]
0 = ‘Non Fraud’	0.98	0.97	0.98	
1 = ‘Fraud’	0.83	0.89	0.86	
Average	0.83	0.89	0.86	0.96

Test Set				[redacted]
0 = 'Non Fraud'	0.98	0.97	0.98	
1 = 'Fraud'	0.82	0.86	0.84	
Average	0.82	0.86	0.84	0.96

Table 3. Classification Report of Decision Tree Model

Despite its strong performance, the Decision Tree model also has several limitations that must be considered. One key drawback is its tendency to overfit, especially when applied to noisy or complex datasets. While we mitigated this to some extent using basic hyperparameter settings, more extensive hyperparameter tuning could further reduce model variance and improve generalization. Furthermore, it is known to be unstable when there are small changes in the training data which could lead to significantly different tree structures. This sensitivity reduces the model's robustness and consistency in production environments. To overcome these challenges, we proceeded with building a Random Forest Classifier, an ensemble method that combines multiple decision trees to improve stability, accuracy, and resilience to overfitting.

3.1.2 Random Forest Classifier

Following on from the Decision Tree model, we extended our modeling approach using a Random Forest Classifier— an ensemble-based learning method designed to improve both predictive performance and model stability. While based on the fundamental structure of decision trees, random forests address their key limitations, particularly high variance and sensitivity to data fluctuations, through bootstrap aggregating (bagging).

Random Forest model will construct multiple decision trees, which each of them are trained on randomly drawn subsets of training data and features. Every tree independently learns patterns from its unique sample and the final prediction is made through majority voting across all trees. This process generates a set of consensus "if-then" rules that can effectively flag anomalies based on transaction amount, time, location, and other relevant features—offering a more robust and auditable framework for fraud detection. This ensemble approach improves generalization by combining the strength of multiple diverse trees. For example, one tree might focus on large night-time transactions, while other emphasizes on large merchant categories; together they reduce overfitting and capture broader fraud patterns. Even without extensive hyperparameter tuning, the Random Forest Classifier demonstrates strong performance, high accuracy, and the ability to handle complex features interactions.

In this study, the model was trained on the same SMOTE-balanced training dataset used for the decision tree, ensuring fair learning from the minority fraud class. We implemented 100 estimators, meaning that 100 individual decision trees are built. This number is chosen to strike a balance between predictive performance and computational efficiency—providing enough trees to capture complex data patterns and ensure robustness, while avoiding excessive training time or resource consumption.

From Table 4, we could see that the model demonstrated strong and consistent performance across validation and test sets with high accuracy score, indicating that all transactions were correctly classified. The precision scores also suggest that the model is able to predict the fraud transaction with a low false positive rate and correctly identify actual fraud cases (i.e., high recall scores of ~86%). All together, these metrics show that the model is highly reliable in distinguishing fraud while keeping false alarms at a manageable level.

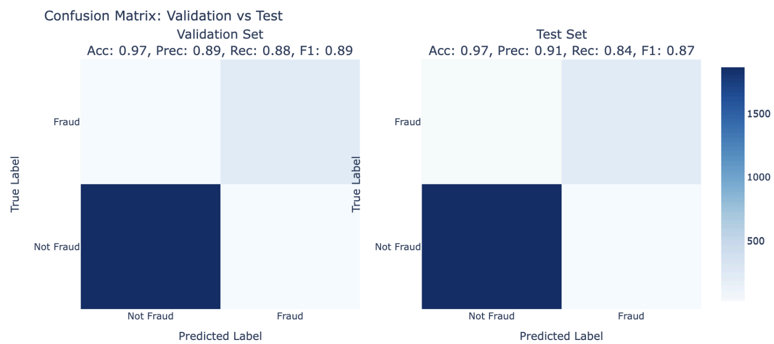


Fig 2. Confusion Matrix of Random Forest Model for Validation and Test Set

Label	Precision	Recall	F1 Score	Accuracy
Validation Set				
0 = ‘Non Fraud’	0.98	0.98	0.98	
1 = ‘Fraud’	0.89	0.88	0.89	
Average	0.89	0.88	0.89	0.97
Test Set				
0 = ‘Non Fraud’	0.98	0.99	0.98	
1 = ‘Fraud’	0.91	0.84	0.87	
Average	0.90	0.84	0.87	0.97

Table 4. Classification Report of Random Forest Model

Despite its strengths, the Random Forest Classifier does come with certain trade-offs, most notably reduced interpretability. Unlike a single Decision Tree model that produces transparent, human-readable “if-then” rules, a Random Forest aggregates predictions from many trees, making it difficult to trace the exact logic behind any single decision. This nature can be a challenge in regulated environments such as banks or financial institutions where transparency is highly required.

However, the model still offers global interpretability through feature importance analysis. This provides valuable insight into which features that most influence the model’s predictions across the dataset. Overall, this model significantly improves upon the decision trees by offering higher predictive accuracy, better generalization to unseen data, and lower risk of overfitting. Its strength in fraud detection lies in its ability to capture complex and non-linear interactions between features while remaining relatively simple to implement. These characteristics make the Random Forest Classifier a strong candidate for real-world deployment, especially in settings where model performance is critical and decisions can be supported by aggregated interpretability tools.

3.1.3 XGBoost

XGBoost is a powerful and scalable implementation of the gradient boosting algorithm that has consistently achieved top performance in many applied machine learning tasks—particularly with structured data such as tabular transaction records. It is known for its high accuracy, built-in regularization, and computational efficiency.

High-level wise, XGBoost builds a sequence of decision trees, where each new tree is trained to correct the errors made by the previous ensemble of trees. This boosting strategy allows the model to focus on misclassified observations, gradually improving prediction accuracy with each iteration. Unlike Random Forests, which build trees in parallel, XGBoost constructs trees sequentially, with each tree designed to improve upon the performance of its predecessors. Unlike Random Forests that build trees in parallel, XGBoost constructs trees sequentially with each tree designed to improve the upon performance of its predecessors.

Beyond standard boosting, this model also includes several enhancements that improve performance and generalizability. In this case, XGBoost uses L1 and L2 regularization to prevent overfitting by penalizing complex models and tree pruning based on the gain thresholds to eliminate any unnecessary branches. Furthermore, the shrinkage (learning rate) will help model to reduce the impact of each tree and slow down the learning process for better generalization. Although early stopping is a supported feature in XGBoost to halt training when validation performance no longer improves, it was not applied in this implementation but could be incorporated in future iterations for improved efficiency.

As with the other supervised models in this study, we trained XGBoost using the SMOTE-balanced training set to address class imbalance, with 100 boosting rounds. We used the default hyperparameters from the Python library to evaluate baseline model performance, set false input for 'use_label_encoder' to avoid deprecation warnings, and specified 'eval_metric=logloss' to ensure appropriate evaluation during training.

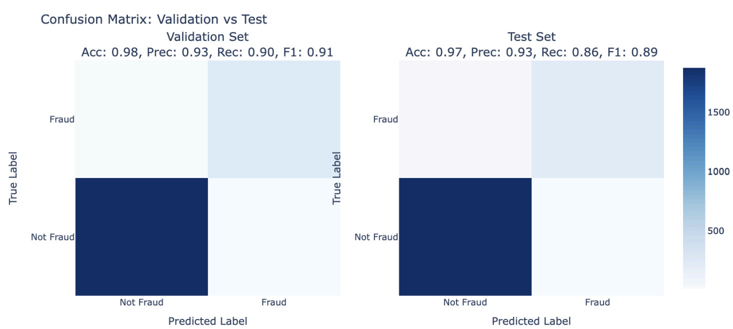


Fig 3. Confusion Matrix of XGBoost Model for Validation and Test Set

Table 5 shows that XGBoost outperformed all other models in the supervised learning as expected with strong performance that remained robust on the both validation and test set. These results demonstrate that XGBoost not only delivers high overall accuracy, but also excels at identifying fraudulent cases while keeping false positives low. This balance is critical in real-world fraud detection, where both missed fraud cases (low recall) and false alarms (low precision) carry substantial costs.

Label	Precision	Recall	F1 Score	Accuracy
Validation Set				[redacted]
0 = 'Non Fraud'	0.98	0.99	0.99	
1 = 'Fraud'	0.93	0.90	0.91	

Average	0.93	0.89	0.91	0.98
Test Set				
0 = ‘Non Fraud’	0.98	0.99	0.99	
1 = ‘Fraud’	0.93	0.86	0.89	
Average	0.93	0.86	0.89	0.97

Table 5. Classification Report of XGBoost Model

Similar to the Random Forest Classifier, XGBoost is not inherently interpretable in the traditional sense. However, the interpretation could be supported using advanced tools such Shapley Additive Explanations (SHAP), which help to explain feature contributions to individual predictions. While XGBoost generally requires greater computational resources and more careful hyperparameter tuning than simpler models, its exceptional predictive performance, support for parallel training, and built-in cross validation make it an ideal model for production-level fraud detection systems, where accuracy and scalability are important.

3.2 Unsupervised Learning: Anomaly Detection Model

In real-world scenarios, fraud patterns continuously evolve due to shifts in market behavior and the emergence of new tactics by fraudsters. These changing patterns often differ significantly from historical behaviors, making it crucial to develop a fraud prevention system capable of detecting novel and previously unseen anomalies. To address this challenge, we implement an unsupervised anomaly detection model, which is well-suited for uncovering hidden patterns that traditional supervised models might overlook.

Since the objective of unsupervised learning itself is to understand the underlying data distribution and discover any hidden patterns that deviate from normal behavior, it is common to not split the dataset into training, validation, and test subsets. This is because no target labels (Y) are available to guide or validate the learning process unlike treatment in supervised learning. Additionally, data splitting could limit the amount of information the model uses to learn the full distribution of normal patterns. This constraint might prevent the model from recognizing rare or subtle anomalies, making it behave more like a supervised model that struggles to generalize beyond known classes.

To maximize the model’s exposure to normal behavior, we train the unsupervised model on the entire dataset after dropping missing values and applying feature scaling. The decision to remove missing values is intentional—by training only on complete, high-quality data, we allow the model to learn well-defined patterns of typical transactions. As a result, incomplete or irregular data points (i.e., data with missing information) are more likely to stand out as anomalies during detection. This approach enhances the model’s ability to flag deviations from established norms, aligning with the core objective of anomaly detection.

Moreover, we do not apply SMOTE or any resampling process since unsupervised models do not rely on predefined classes, so creating synthetic samples might introduce bias and could distort the original structure of the data. We are also unable to perform any validation whether those artificial samples align with the general anomalies or normal patterns, thus applying SMOTE may unintentionally hinder models’ ability to detect meaningful deviations.

Overall, unsupervised learning offers a valuable approach for detecting emerging and unknown fraud behaviors, especially in dynamic environments where historical labels are limited or may quickly become outdated.

3.2.1 Isolation Forest

Isolation Forest is an unsupervised machine learning algorithm specifically designed for anomaly detection. Adapting from the structure of random forests, the algorithm works by randomly selecting a feature and then randomly choosing a split value between the feature's minimum and maximum values. This process is repeated recursively to build an ensemble of isolation trees. Furthermore, the unique feature of the Isolation Forest model is that it focuses on the isolation rather than profiling of data points. Anomalies are generally isolated more quickly than normal points as they tend to differ significantly from the rest of the data and require fewer random splits to separate. This resulted in the shorter average path length from the root node in the trees, which the model uses to assign anomaly scores.

Unlike distance-based methods like k-Nearest Neighbors (k-NN) or Local Outlier Factor (LOF), Isolation Forest does not rely on pairwise distance calculations. This makes it particularly suitable for high-dimensional data, as it is less susceptible to the curse of dimensionality. Additionally, as a tree-based method, Isolation Forest is invariant to monotonic transformations of features and does not require feature scaling or normalization, although preprocessing may still improve interpretability and performance. Benefitting from its computational efficiency, low memory requirement, and strong performance in handling complex and high-dimensional datasets, Isolation Forest model is highly effective and practical for identifying fraudulent behavior in real-world data.

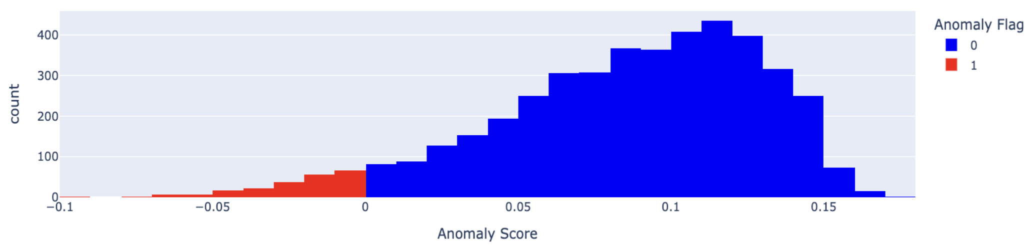


Fig 4. Anomaly Score Distribution of Isolation Forest Model

As shown in Figure 4, the anomaly score distribution is derived from the Isolation Forest model using 100 estimators, meaning 100 trees in the ensemble. A higher number of estimators generally leads to more stable and reliable anomaly scores, as the ensemble better captures the underlying data structure. We assume that approximately 5% of the data points are anomalous, reflecting a typical scenario in fraud detection where anomalies are rare.

The distribution indicates that potential anomalies (i.e., flagged in red bars) are concentrated to the left side of the score distribution. This is because the Isolation Forest's built-in function of *'.decision_function()'* will assign lower scores to more anomalous observations — with negative scores signaling a higher likelihood that those transactions are fraud. As a result, red bars cluster toward the lower end of the distribution.

Moreover, the red bars are significantly fewer than the blue bars (representing normal transactions), which confirms both the imbalance in the dataset and the model's assumption that anomalies are statistically distinct and infrequent. This aligns well with real-world fraud detection scenarios, where fraudulent transactions typically form a small, isolated minority that deviates from the patterns of normal behavior.

Label	Precision	Recall	F1 Score	Accuracy
All dataset				
0 = ‘Non Fraud’	0.87	0.98	0.92	
1 = ‘Fraud’	0.74	0.23	0.35	
Average	0.85	0.86	0.83	0.87

Table 6. Classification Report of Isolation Forest Model

Although standard supervised evaluation metrics are not typically applicable to unsupervised models due to the absence of labeled data, we chose to compute these metrics using the original target variable for internal assessment and comparison purposes. This approach helps evaluate how well the unsupervised model can detect anomalies relative to known fraud cases, even if it was not trained on the labels. As shown in Table 6, the model achieves high precision for the fraud class, indicating that it correctly flags transactions as anomalous. However, the model suffers from low recall issues, meaning it misses a significant number of actual fraud cases. This is expected given the highly imbalanced dataset and the lack of label guidance during training.

Furthermore, since the Isolation Forest relies solely on structural deviations within the data rather than labeled outcomes, it may fail to detect subtle anomalies that have similar behavior with normal patterns but are indeed fraudulent. Conversely, it might also isolate rare but benign patterns, flagging them as potential fraud despite being legitimate. This trade-off highlights the inherent limitation of unsupervised approaches and underscores the need to interpret results with caution, especially in critical applications like fraud detection.

3.2.1 Auto Encoders

Autoencoders are a type of neural network model designed to learn compressed representations of input data by minimizing the reconstruction error. This capability makes them particularly effective for capturing the underlying structure of data and uncovering unusual patterns—such as those found in fraudulent transactions. The model operation is composed in two main stages: Encoding process, where input data is transformed into a lower-dimensional latent space representation, and Decoding process, where the compressed representation is reconstructed back to its original form.

Because the model is trained on normal data, it becomes proficient at reconstructing typical patterns. In consequence, any data points that deviate significantly from these patterns such as anomalies or fraud, are reconstructed poorly, resulting in high reconstruction errors. Data points with errors exceeding a users’ predefined threshold are flagged as anomalies.

What makes Autoencoders well-suited for anomaly detection is due to their interpretability, which the reconstruction error itself can be used as an anomaly score. Their ability to operate in high-dimensional feature spaces (e.g., time series or tabular data) allows them to capture complex interactions, and the lower-dimensional latent space helps reduce storage needs while preserving critical information. Additionally, Autoencoders require minimal preprocessing which typically just requires data to be normalized and do not need labeled data, hence making them ideal for real-world fraud detection where labeled anomalies are rare and evolving.

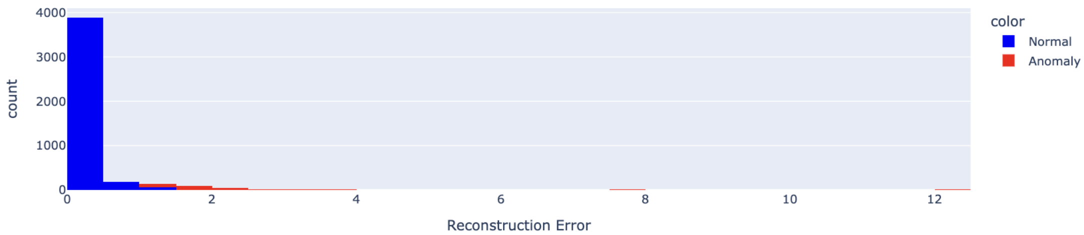


Fig 5. Reconstruction Error Score Distribution of Auto Encoders Model

Label	Precision	Recall	F1 Score	Accuracy
All dataset				
0 = 'Non Fraud'	0.88	1.00	0.94	
1 = 'Fraud'	0.93	0.29	0.45	
Average	0.89	0.88	0.86	0.88

Table 7. Classification Report of Auto Encoders Model

Based on Figure 5, the distribution of reconstruction errors shows that normal transactions are concentrated on the left, indicating low reconstruction error. This aligns with expectations, as the model is trained to accurately reconstruct typical patterns found in normal data. Meanwhile, anomalous transactions appear on the right, associated with higher reconstruction errors. However, the presence is very limited which may indicate that the model struggles to detect many fraudulent cases, reflected in the low recall score (Table 7). This limitation stems from the nature of Autoencoders, which are optimized to reconstruct normal data, not necessarily to differentiate anomalies explicitly. As a result, the model may fail to detect subtle deviations if they do not significantly differ from normal patterns.

Furthermore, model performance is highly sensitive to hyperparameter tuning, especially the threshold used to define anomalies. A higher threshold can improve precision (fewer false positives) but reduce recall (more missed anomalies), while a lower threshold may improve recall but lead to more false positives. Another limitation from this model is that Autoencoders can overfit to the normal data structure, causing them to reconstruct even slightly abnormal inputs well and misclassify them as normal data. This overfitting issues would lead to false negatives, where rare or potentially fraudulent transactions are not flagged, despite being unusual in nature.

3.3 Feature Importance Analysis

As part of the integrated fraud detection analysis framework, we perform feature importance analysis to identify which features have the most significant impact on the model's output. These insights help users focus on key drivers when developing effective fraud prevention strategies. Additionally, this analysis enhances the transparency of supervised models by offering explanations behind the model's predictions. Understanding feature importance also supports better model tuning in a way that it helps users to adjust thresholds or decision rules on high-impact variables. Moreover, it can guide dimensionality reduction by identifying and retaining only the most relevant features, which reduces noise, mitigates overfitting, and can lead to improved model performance.

To compute importance scores, we use the scaled training dataset for supervised models (i.e., Decision Tree, Random Forest, and XGBoost), and the entire scaled dataset for unsupervised models (i.e., Isolation Forest and Autoencoders).

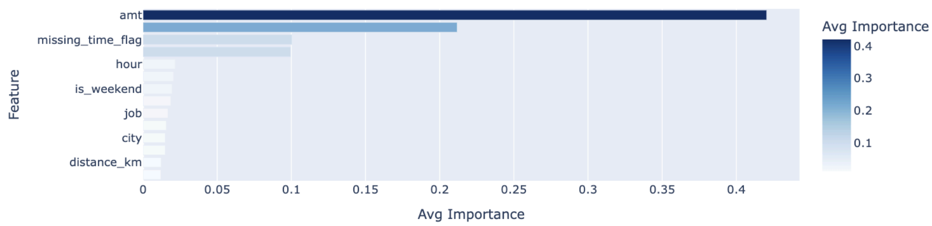


Fig 6. Features Importance Score for Supervised Learning Models

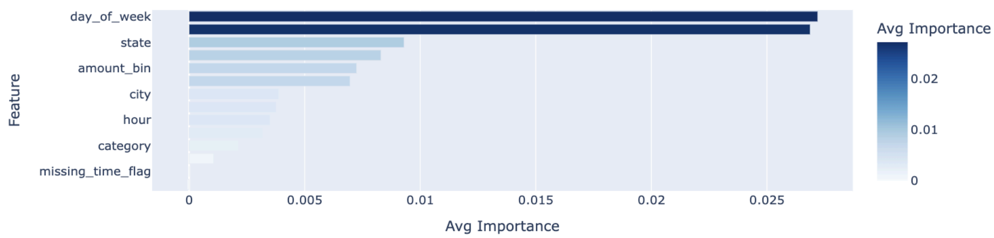


Fig 7. Features Importance Score for Unsupervised Learning Models

Both Figure 6 and Figure 7 show the average importance scores from all models under each supervised and unsupervised learning. The transaction amount feature (i.e., amount and amount bin category) is consistently being as the important features in both model, which followed by the transaction time features such as hour, missing_time_flag (i.e., indicate the missing transaction time information), and day of the week (i.e., Monday). To understand deeper on the features importance from each model implemented, we also visualise separate importance score distribution for each model (Figure 8).

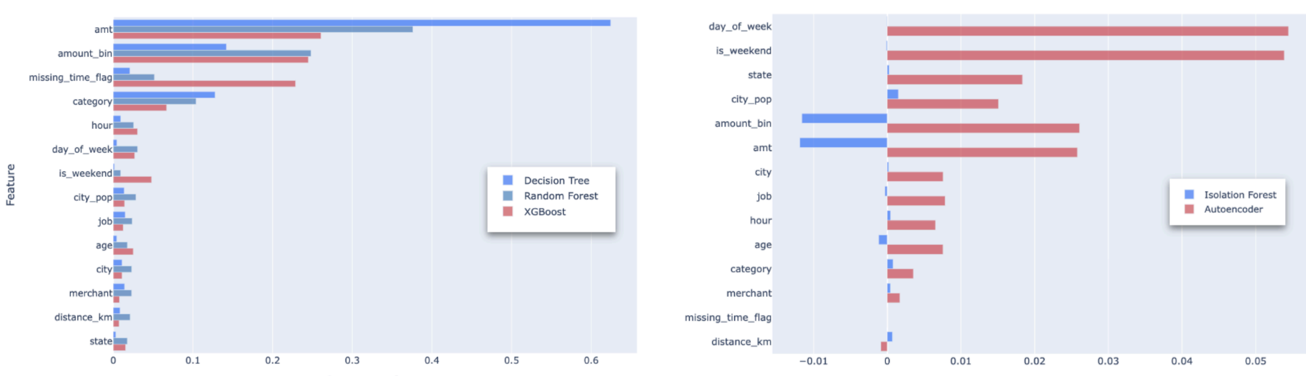


Fig 8. Features Importance Score for Supervised Learning (Left) and Unsupervised Learning Models (Right)

For supervised learning models, the transaction amount features (i.e., amount and amount bin) consistently emerged as the most influential across all three models, suggesting their central role in distinguishing fraudulent from legitimate transactions. Notably, only XGBoost identified the missing time flag feature as key predictor, with Random Forest assigning it a moderate importance score. This suggests that more sophisticated ensemble methods (i.e., Random Forest and XGBoost) are capable of detecting nuanced signals, such as the absence of timestamp information, as potential fraud indicators.

From the reliability perspective, those ensemble methods also exhibit more stable and interpretable results than single Decision Trees. As it can be seen from the left side of Figure 8, the latter model shows sharp spikes in feature importance that is likely due to overfitting to particular splits, which this issue is mitigated by ensemble methods through averaging results across many trees. This aggregation captures both individual and interaction effects between features and helps avoid overfitting, making their importance scores more robust and generalizable.

In contrast, unsupervised learning models show notable differences in feature importance. As shown in the right side of Figure 8, the Autoencoders assigns a positive importance score to the transaction amount feature, while the Isolation Forest assigns it a negative score. This divergence stems from the underlying algorithms and scoring methods adapted by each model.

Autoencoders learns to reconstruct normal patterns and flag anomalies via reconstruction error. Features that consistently cause higher reconstruction errors are deemed more important, hence resulting in positive importance scores, which are typically calculated from the absolute reconstruction loss per feature. On the other hand, Isolation Forest isolates anomalies through random feature splits in A feature is considered important if it helps isolate outliers with shorter paths (i.e., fewer splits). If a feature like transaction amount requires deeper trees to isolate anomalies, it reduces the model's efficiency and is assigned a lower or negative relative importance. This means that while a feature might strongly influence reconstruction loss (as in Autoencoders' case), it may not be efficient in isolating anomalies within a tree-based structure.

The difference in model algorithms itself highlights how the choice of model would affect the interpretation of feature importance. Thus, combining results from both supervised and unsupervised models offers a more holistic understanding of which features are most critical in fraud detection, hence enabling decision makers to assign appropriate weight to the most impactful drivers.

4. Risk Scoring Model

Instead relying on the fraud label prediction, it is important to complement the prediction results with quantitative and interpretable measures to show the likelihood of a transaction to be fraudulent. Therefore, we perform a risk scoring model to provide more granular insight for users in the real world practical case. The final product of risk scores allows them to prioritize further investigations on high-risk transactions. It can be integrated to the fraud monitoring system, allowing institutions to execute immediate action (e.g., halt transactions) based on the scores generated and eventually improve the model transparency to stakeholders and regulators to fulfill the compliance requirements. There are two models that are being performed in this step, which are model-based that applies the supervised learning model and rule-based method that generated from users defined input.

4.1 Model-Based Method

In this part, we utilized the trained supervised learning techniques to estimate the fraud likelihood by constructing model-based risk score models. Specifically, we used the trained Random Forest and XGBoost Classifier (in Part 1) to predict the probability of each transaction being fraudulent, producing scores ranging from 0 to 1. To interpret those probabilities, we categorized three risk levels: (1) High risk for probability above 0.9, (2) Medium risk for probability between 0.6 and 0.9, and (3) Low risk for probability under 0.6. These thresholds were chosen based on the distribution of predicted probabilities, where transactions with scores above 0.9 were strongly indicative of fraud. Meanwhile, those below 0.6 typically represented normal behavior and values in between captured uncertain or borderline cases.

Risk Level Category // Counts	Random Forest	XGBoost
High	3,272	13,578
Medium	8,942	523
Low	2,232	345

Table 8. Risk Score Distribution of Model-Based Method (Data without SMOTE)

Initially, when applying the scoring method into our dataset (scaled without resampling), our model-based results revealed discrepancies in risk categorization largely due to the inherent class imbalance dataset (i.e., only 12.77 % of transactions were labeled as fraud). This imbalance led to inconsistent risk scoring and over flagging by models. As shown from Table 8, XGBoost model classified a significant disproportional amount of transactions as high risk, while Random Forest model labeled the most for medium risk category. These misalignment patterns suggest that both models, particularly XGBoost, may overestimate the fraud likelihood due to their sensitivity to the minority class. Consequently, both models tend to underrepresent the dominant class (non-fraud) and struggle to capture more nuanced and complex fraud patterns effectively.

To address the misalignment issues caused by class imbalance, we applied the SMOTE (Synthetic Minority Over-sampling Technique) method during data preprocessing. SMOTE was used after feature standardization and train-test splitting to generate synthetic samples for the minority (fraud) class, thereby balancing the dataset and promoting more stable predictions.

Model	Risk Level Category // Counts	Validation Set	Test Set
Random Forest	High	181	178
	Medium	82	70
	Low	1,904	1,919
XGBoost	High	241	234
	Medium	21	17
	Low	1,905	1,919

Table 9. Risk Score Distribution of Model-Based Method (Data with SMOTE)

After performing the resampling process, it can be seen from Table xx that both models show consistency results across all risk level categories in each validation and test set. The false positives (i.e., where model assigns high risk scores of 0.9 for non fraud data with label = 0) dropped from 11,897 to 10 for XGBoost model and from 2,660 to 8 for Random Forest model. These results indicate a significant improvement in model performance, particularly in distinguishing fraudulent from non-fraudulent transactions more accurately.

4.1.1 Confusion Matrix Analysis for Supervised Learning

Building on the model improvements, we conducted a confusion matrix analysis for both the Random Forest and XGBoost classifiers. The matrix itself maps the predictions into the original class which data is referred to, hence it is only applicable when the distribution of predicted variables (i.e., fraud label class) is known. Therefore, this analysis is crucial in evaluating the performance of supervised learning models that perform a classifier role in the fraud detection tasks.

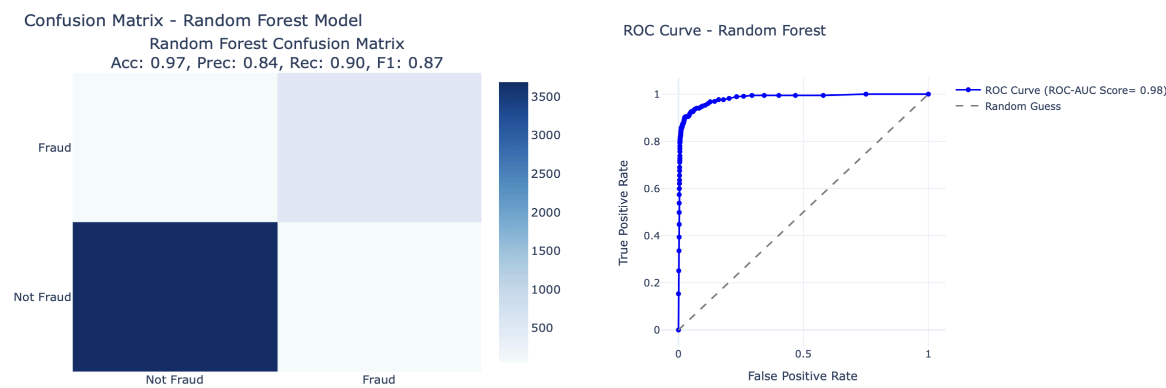


Fig 9. Confusion Matrix and ROC Curve of Random Forest Model

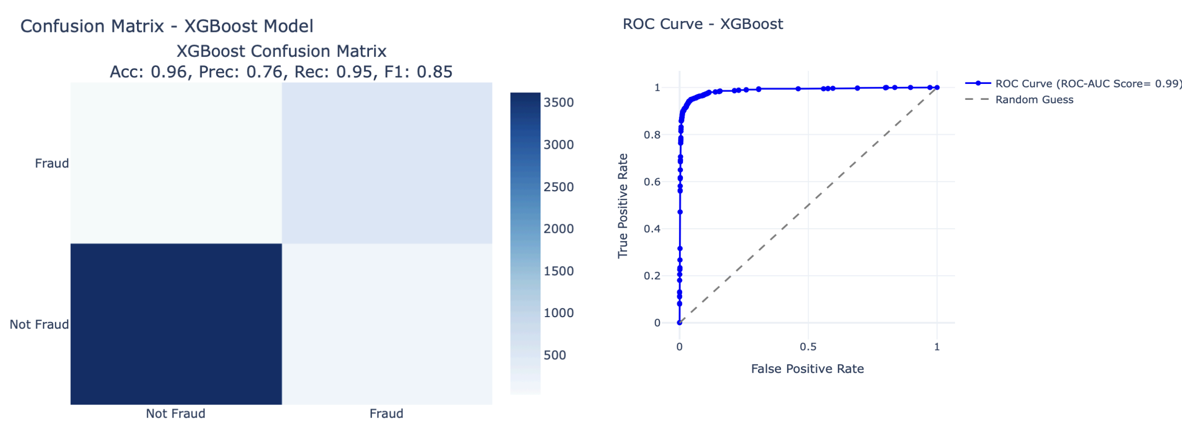


Fig 10. Confusion Matrix and ROC Curve of XGBoost Model

Based on Figure 9 and 10, both Random Forest and XGBoost models demonstrated strong overall performance with accuracy rates exceeding 96%. This implies that the model is correctly classified 96% out of samples in the combined validation and test set. However, beyond accuracy, we need to look at other metrics that provide more nuanced evaluation on the models' effectiveness in making predictions.

The Random Forest model strikes a solid balance between high recall and low false positive rate, reflected in a precision of 84% and a recall of 90%. This indicates that the model not only detects the majority of fraud cases but also avoids misclassifying many legitimate transactions. Such balance makes Random Forest a reliable and conservative choice in fraud detection. In contrast, the XGBoost model, despite a lower precision of 76%, achieves a remarkably high recall of 95%, meaning it identifies nearly all actual fraud cases. This suggests that XGBoost adopts a more aggressive approach in flagging potential fraud, which can be especially valuable in environments where the cost of missing a fraudulent transaction outweighs the inconvenience of flagging legitimate ones.

Overall, these results show that Random Forest is more cautious by minimizing false positives and preserving customer experience, while XGBoost is more aggressive by maximizing fraud detection even at the cost of more false alarms. Therefore, the optimal model choice depends on the risk tolerance and operational priorities of the fraud detection system. In scenarios where missing fraud is highly costly, XGBoost may be preferable. Conversely, if the primary concern is avoiding false accusations of legitimate users, Random Forest offers a more favorable trade-off.

4.2 Rule-Based Method

Following the model-based approach, we developed a complementary rule-based risk scoring model to enhance and support the machine learning-based fraud detection system. This model assesses fraud risk on a 10-point scale, with thresholds defined as: (1) Low risk (<4 points), (2) Medium risk (4–6 points), and (3) High risk (>6 points). The scoring logic is derived from our earlier Exploratory Data Analysis (EDA), which identified Transaction Amount and Merchant Category as key variables strongly correlated with fraud. As a result, we assigned higher weights of 3-6 points to these riskier factors — such that: (1) if a transaction amounts beyond $500 or $1,000; or (2) the merchant is an online shop or offline groceries.

In addition, we incorporated secondary risk indicators such as transaction distance, timestamp anomalies, and cardholder age outlier. Each of these factors contributes 1 point to the score to reflect their relevance without allowing them to dominate the overall assessment. As shown from Figure 11, the resulting risk score distribution indicates that most transactions were classified as low risk, while 773 transitions were flagged as high risk. This result aligns with the overall dataset's class imbalance, suggesting rule-based thresholds are effective for initial risk classification.

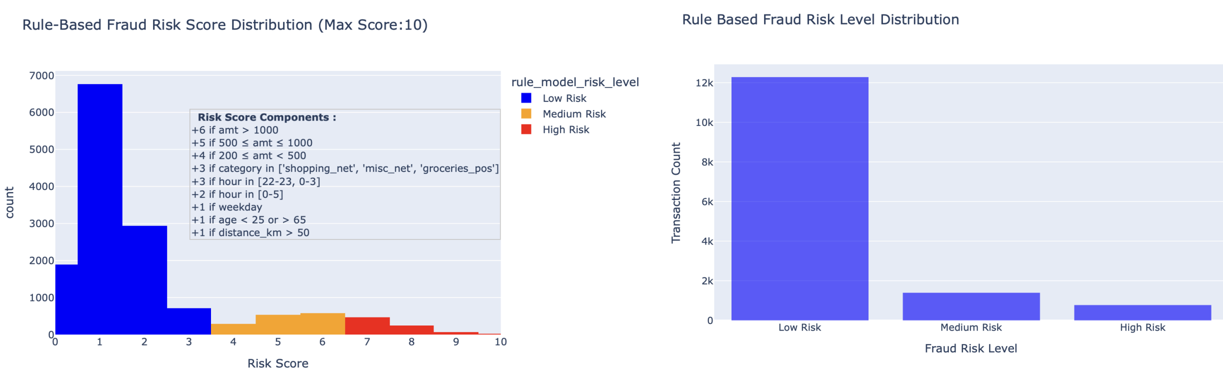


Fig 11. Risk Score and Risk Level Distribution of Rule-Based Model

However, the rule-based model has inherent limitations as it relies heavily on expert judgement and static rules derived from EDA, which may not generalize well to unseen or evolving fraud patterns. Since the factor weights are fixed, the model may need further manual adjustment when it comes to new data. Therefore, while it is useful as a complementary tool, the rule-based model is best employed in conjunction with the model-based approach to ensure adaptability and robustness in a real- world dynamic fraud detection environment.

4.3 Hybrid (Rule-Enhanced) Based Method

To leverage both data-driven accuracy (model-based method) and expert insight (rule-based method), we developed a hybrid risk scoring model that combines Random Forest predictions with domain expert rules. This approach retains the interpretability of rule-based systems while enhancing robustness through machine learning. We first trained a Random Forest classifier to generate a base fraud probability score for each transaction. This score was then adjusted based on risk-enhancing conditions (Table 10), guided by feature importance analysis with total score adjustments capped at 1 to maintain calibration and avoid excessive inflation. Eventually, the final hybrid score aims to balance model precision with human-interpretable logic, enabling more reliable and explainable fraud detection.

Risk Indicator	Transaction Characteristics	Score adjustment
High	Large transaction volume Risky merchant categories	+0.2 points
Medium	Transaction time between 0 am to 3 am	+0.1points
Low	Distance > 100km Unusual timestamps	+0.05 points

Table 10. Risk Score Adjustment Rules for Hybrid Model

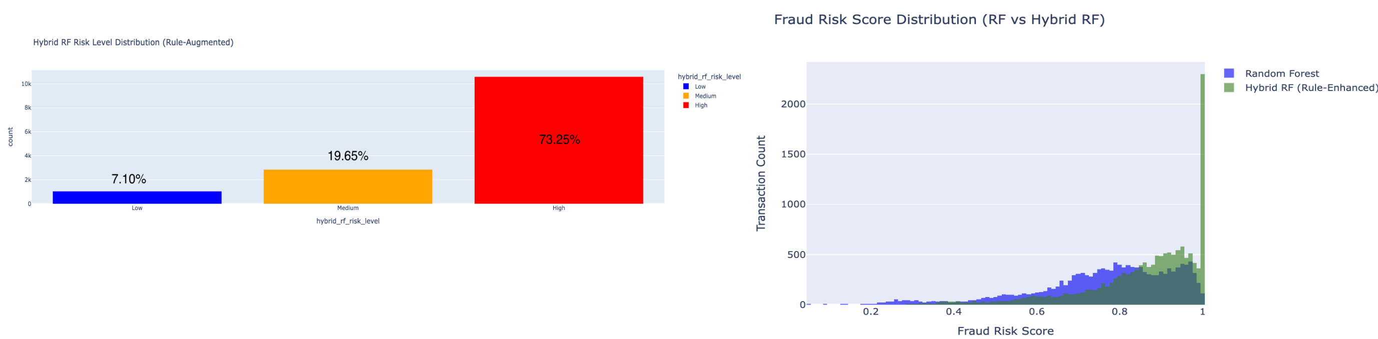


Fig 12. Risk Level Distribution of Hybrid Based Model

From the left chart of Figure 12, we observe that the hybrid model classifies most transactions as high risk (i.e., 73% of transactions), demonstrating increased sensitivity to risk signals and an ability to capture a wider range of potentially fraudulent activities. The risk score distribution, on the right, (Figure 12) shows a pronounced right-skew under the hybrid approach, reflecting a significant increase in transactions assigned high risk scores. By contrast, the standalone Random Forest model presents a broader, more gradual distribution, especially across the medium-risk range. The hybrid model, however, shifts this distribution, creating a clearer separation between risky and non-risky transactions. This enhanced differentiation indicates that the hybrid method amplifies risk sensitivity, enabling more decisive high-risk flagging and reducing classification ambiguity.

4.3.1 Confusion Matrix Analysis for Hybrid (Rule-Enhanced) Model

To evaluate the effectiveness of our fraud detection approach, we compared the performance of two models: random forest classifier model, and hybrid risk scoring model, which enhances the random forest classifier predictions using a set of rule-based adjustments derived from domain expertise. Random forest classifier model serves as the baseline of supervised learning approach in our study.

As shown in Figure 13, the Random Forest model delivers strong performance with high accuracy scores of 97%, 84% precision. Additionally, the area under the ROC curve (AUC) is 0.99, reflecting excellent discrimination between fraudulent and non-fraudulent transactions. These metrics confirm the Random Forest's capability to effectively detect fraud. However, a notable limitation of this model is its lack of interpretability. It provides limited transparency in explaining why a particular transaction is classified as high or low risk, which can hinder trust and usability in operational settings.

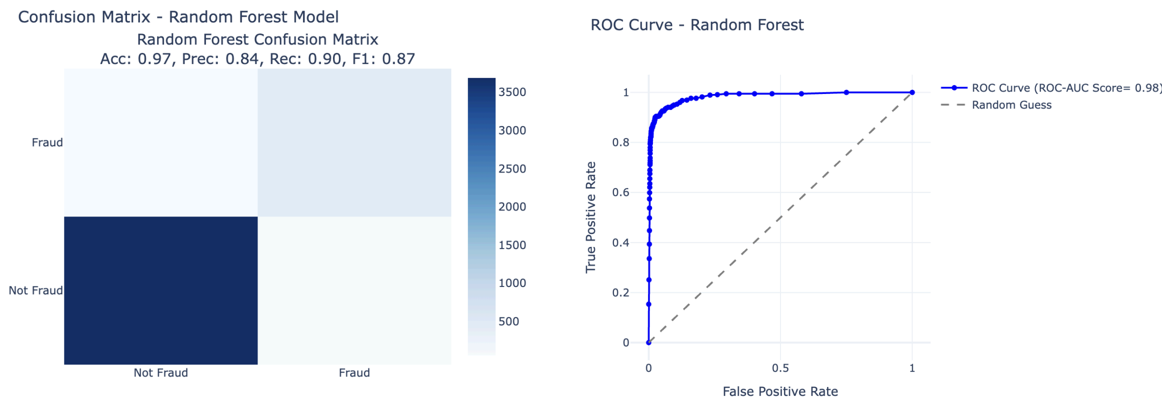


Fig 13. Confusion Matrix and ROC Curve of Random Forest Model

Furthermore, compared with Figure 14, the Hybrid approach reflects a clear trade-off. Its overall accuracy drops to 81%, with an F1 score of 0.34 and an AUC of 0.58, compared to the stronger statistical performance of the Random Forest model. This decline is expected, as the hybrid model incorporates rule-based overrides that prioritize interpretability and control over maximizing pure predictive accuracy.

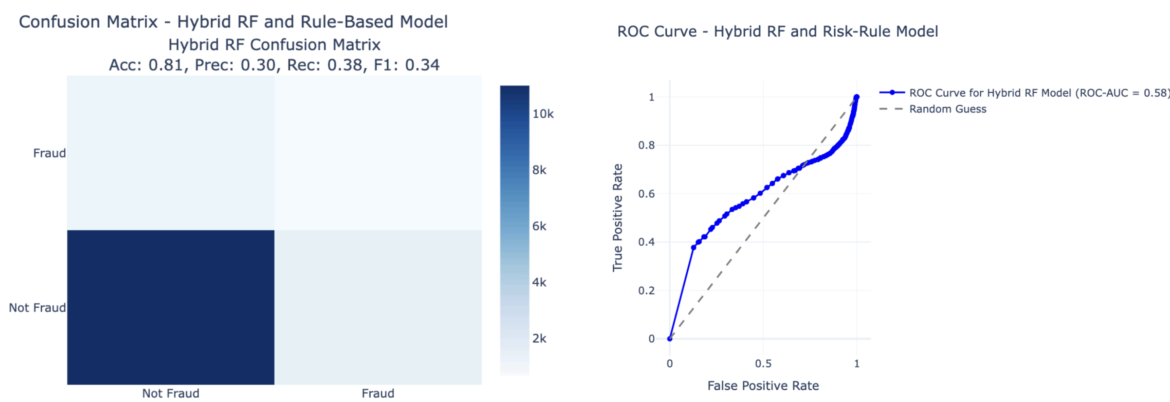


Fig 14. Confusion Matrix and ROC Curve of Hybrid-Based Model

Despite the declining performance metrics, the Hybrid model offers a key operational advantage when focusing on the top-risk transactions. By analyzing the top 10% of transactions flagged as high risk from each model, we found that the Random Forest model correctly identified 19.58% of actual fraud cases, whereas the Hybrid model improved this proportion to 30.26%. Through the integration of domain knowledge into the scoring process, the hybrid model enhances the prioritization of truly high-risk cases, even if its overall aggregate metrics are lower.

5. Conclusion

Aimed to develop an effective fraud detection framework, we combined both supervised learning models and unsupervised anomaly detection techniques to generate the best outcome. For the supervised approach, we implemented Decision Tree, Random Forest Classifier, and XGBoost on SMOTE-balanced dataset to address class imbalance. Among these, XGBoost consistently outperformed others with a precision of 0.93, recall of 0.86, and accuracy of 0.97 on the test set, making it the most effective at identifying fraudulent transactions while minimizing false positives. To improve interpretability and practicality, we designed a hybrid risk scoring model that combines Random Forest outputs with domain expert rules. This gives us a more robust result and explainable fraud alerts.

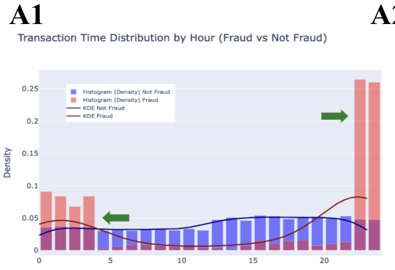
Beyond supervised models, we explored unsupervised models – Isolation Forest and Auto Encoders to detect anomalies without relying on labeled data. While these models provided less accuracy precision score compared to supervised learning models, they offered complementary insights for identifying anomalous fraud behavior outside the training dataset. This dual modeling strategy provides insights for, namely, financial institutions with a layered approach of leveraging historical fraud knowledge via supervised learning and staying adaptive through unsupervised detection.

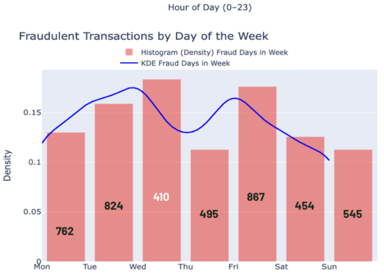
Furthermore, to support operational decision-making, we developed a comprehensive risk scoring framework comprising three layers: (1) Model-based method using predicted fraud probabilities, (2) Rule-based method informed by domain expertise, and (3) Hybrid approach that adjusts model predictions based on risk-enhancing conditions. While the Hybrid model sacrificed some statistical performance, it significantly improved the capture rate of high-risk cases, detecting 30.26% of actual frauds in the top 10% flagged transactions in comparison to 19.58% from the Random Forest Model alone; enabling more efficient fraud investigation efforts.

Despite its promising outcomes, the models built for this study have several limitations which can be adopted for future improvement. Applying hyperparameter tuning could further enhance models' performance, especially for models like XGBoost and Random Forest Classifier, whose predictive power and generalization strongly depend on tuning parameters. Incorporating that technique such as grid search could improve fraud detection accuracy and reduce false positives by optimizing model complexity and sensitivity. Furthermore, the hybrid model rule-based component relies on a limited number of manually defined conditions. While these rules capture known fraud behaviors, they can be adaptable. Expanding the rule set using leveraged techniques like automatic rule learning could make the hybrid model more responsive and data-driven, reducing the dependence on predefined heuristics and allow continuous updates based on fraud trends. Finally, future work could explore alternative models such as LightGBM or Deep Neural Networks (DNN) to improve efficiency, scalability, and detection capabilities in real-world deployment scenarios.

6. Appendix

A1





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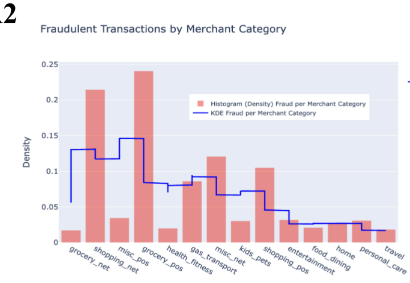

merchant_category	Total_Transactions_Per_Cat	Fraud_Transactions_Per_Cat	Fraud_Rate
grocery_pos	1602	444	28.0
shopping_net	1408	396	28.0
misc_net	821	223	27.0
shopping_pos	1354	194	14.0
gas_transport	1430	159	11.0
travel	385	34	9.0
misc_pos	823	64	8.0
grocery_net	474	32	7.0
entertainment	953	59	6.0
personal_care	990	57	6.0
kids_pets	1141	56	5.0
food_dining	870	39	4.0
health_fitness	891	37	4.0
home	1304	51	4.0

A3